Assignment 5

NIS3352-01 Modern Cryptography(2)

Marking Rubrics for Assignments.

- You will receive full marks if you idea is (mostly) correct and you haven showed an adequate effort to present your ideas. No marks will be deducted if there are small errors or typos.
- If you are missing an important point of a problem, yet you have partially answered it, 10 marks of the question will be deducted.
- Full marks of the problem will be deducted if (1) your idea is not correct at all or (2) there is an obvious sign that you are copying it from your classmates.
- Each assignment is given one week to finish. Yet, we still accept it if it is at most one week late. For assignments two weeks after the deadline, 10 marks will be deducted.

Problem 1 (25 marks) Let $A \in \mathbb{Z}_q^{n\times m}$ and define a hash function

$$H_A(x) := Ax \bmod q, \qquad x \in \{0,1\}^m.$$

Prove that if one can find a collision for H_A, then one can construct a solution to the SIS problem.

Problem 2 (25 marks)

The Inhomogeneous Short Integer Solution (ISIS) Problem. Let $q \geq 2$, and let

$$A \in \mathbb{Z}_q^{n\times m}, \qquad u \in \mathbb{Z}_q^n.$$

The *Inhomogeneous Short Integer Solution* problem, denoted $\mathrm{ISIS}_{n,m,q,\beta}$, asks to find a vector

$$x \in \mathbb{Z}^m$$

such that

$$Ax \equiv u \pmod q$$

and

$$\|x\| \leq \beta.$$

In other words, ISIS asks for a short preimage of a given target syndrome u under the linear map defined by A modulo q. Note that SIS is the homogeneous special case of ISIS corresponding to the target $u = 0$.

A Hash-and-Sign Scheme Based on ISIS. Let

$$H : \{0,1\}^* \to \mathbb{Z}_q^n$$

be a hash function, modeled as a random oracle. Consider the following idealized signature scheme.

- **Public key:** a matrix

$$A \in \mathbb{Z}_q^{n\times m}.$$

- **Secret key:** a trapdoor algorithm $\mathcal{O}_A$ which, on input any

$$u \in \mathbb{Z}_q^n,$$

outputs a short vector

$$\sigma \in \mathbb{Z}^m$$

such that

$$A\sigma \equiv u \pmod q, \qquad \|\sigma\| \leq \beta.$$

- **Signing algorithm:** on input a message $\mu \in \{0,1\}^*$, compute

$$u := H(\mu) \in \mathbb{Z}_q^n,$$

and output

$$\sigma \leftarrow \mathcal{O}_A(u).$$

- **Verification algorithm:** on input a message-signature pair (μ, σ), accept if and only if

$$A\sigma \equiv H(\mu) \pmod q \qquad \text{and} \qquad \|\sigma\| \leq \beta.$$

Prove that the above signature scheme is existentially unforgeable under chosen-message attacks (EUF-CMA) in the random oracle model, assuming that the corresponding $\mathrm{ISIS}_{n,m,q,\beta}$ problem is hard.

More precisely, show that if there exists a probabilistic polynomial-time adversary $\mathcal{A}$ that wins the EUF-CMA game against the above signature scheme with non-negligible probability, then one can construct a probabilistic polynomial-time reduction $\mathcal{B}$ that solves the $\mathrm{ISIS}_{n,m,q,\beta}$ problem with non-negligible probability.

Problem 3 (25 marks)

1. Let $f(x) = x^2 + 1 \in \mathbb{F}_3[x]$.
   (a) Prove that $f(x)$ is irreducible over $\mathbb{F}_3$.
   (b) Construct the field
   $$\mathbb{F}_9 \cong \mathbb{F}_3[x]/(x^2+1).$$
   If $\alpha = [x]$, express every element of $\mathbb{F}_9$ in the form $a + b\alpha$ with $a, b \in \mathbb{F}_3$.
   (c) Find the multiplicative inverse of every nonzero element of $\mathbb{F}_9$.
   (d) Determine all generators of the cyclic group $\mathbb{F}_9^\times$.

2. Let
   $$\mathbb{F}_8 \cong \mathbb{F}_2[x]/(x^3+x+1),$$
   and let $\alpha = [x]$.
   (a) Compute
   $$\alpha^3,\ \alpha^4,\ \alpha^5,\ \alpha^6,\ \alpha^7.$$
   (b) Use these computations to determine the multiplicative order of α.

3. Let F_q be a finite field with q elements. Prove that the multiplicative group
   $$F_q^\times = F_q \setminus \{0\}$$
   is cyclic.

4. Let $\mathbb{F}_{p^n}$ be a finite field, and let d be a positive integer. Prove that $\mathbb{F}_{p^n}$ contains a subfield of size p^d if and only if $d \mid n$.

5. List all possible sizes of subfields of $\mathbb{F}_{2^{16}}$.

Problem 4 (25 marks)

1. Let C be the binary linear code with generator matrix

$$G = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \end{pmatrix}.$$

   Answer the following questions:

   (a) List all codewords of C.

   (b) Determine the parameters $[n, k, d]$ of C.

   (c) Find a parity-check matrix H for C.

   (d) Suppose the received word is

$$r = (1, 1, 0, 1, 1).$$

   Compute its syndrome Hr^T.

2. Let $C \subseteq \mathbb{F}_3^4$ be the linear code generated by

$$G = \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 1 \end{pmatrix}.$$

   Answer the following questions:

   (a) Determine $|C|$.

   (b) List all codewords of C.

   (c) Compute the minimum distance of C.

   (d) Decide whether C can correct one error.

3. Let C be an $[n, k, d]$ linear code over $\mathbb{F}_q$. Prove the Singleton bound:

$$d \leq n - k + 1.$$

4. Let C be the binary linear code generated by

$$G = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \end{pmatrix}.$$

   (a) Use elementary row operations to transform G into a systematic generator matrix of the form

$$G' = (I_k \mid P).$$

   (b) Write down the corresponding parity-check matrix

$$H = (-P^T \mid I_{n-k}).$$

   (c) Verify that

$$G'(H)^T = 0.$$